1 A thin copper sheet is suspended from a small hole near the top of the sheet and placed in a magnetic field, as shown in Fig. 1.1.

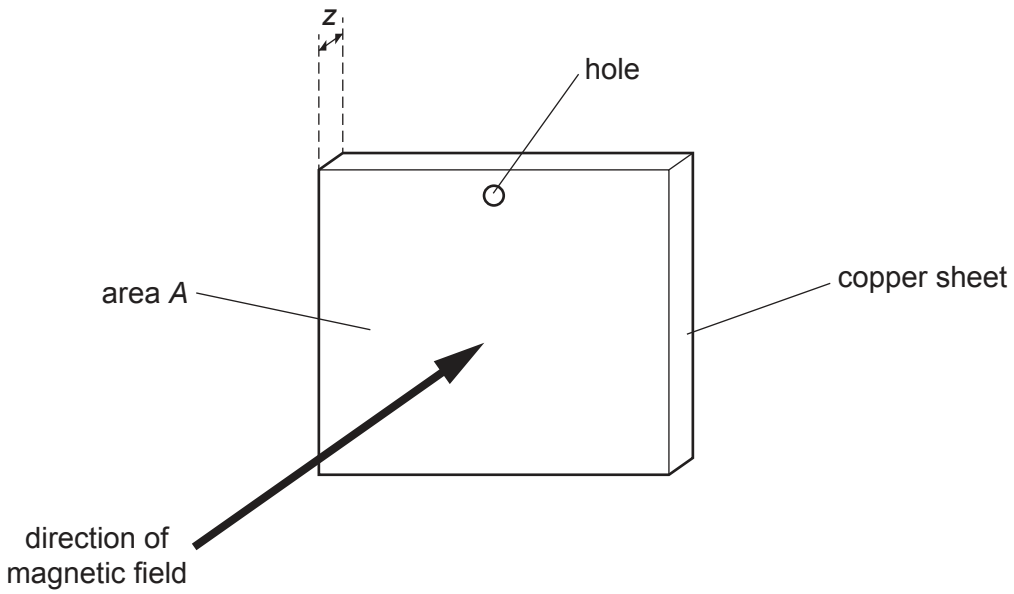

Fig. 1.1 (not to scale)

The sheet has area A and thickness z.

The sheet is displaced from its equilibrium position and then released so that it oscillates perpendicular to the direction of the magnetic field. The time t from when the sheet is released to when it becomes stationary is measured.

It is suggested that t is related to z by the relationship

$$t = \frac{Kz^q}{AB\rho}$$

where B is the magnetic flux density of the field, ρ is the density of copper, and K and q are constants.

Plan a laboratory experiment to test the relationship between t and z.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and q.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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